

Literature Review

Image Encryption Based on Textural Feature Analysis

Senior Cryptography Research Analysis

Papers Reviewed: 7 (Paper 3 excluded - empty file)

Prepared for: Thesis Documentation

Topic: 3-Tier Adaptive Chaotic Image Encryption System

Paper 1: Image Encryption Using Chaotic Maps - A Comparative Cryptanalysis

Authors: A. Dinu & M. Frunzete (2025)

Journal: *Mathematics*, 13, 2588

1. Research Objective

To systematically evaluate and compare classical chaotic maps (Henon, Tent, Logistic) against AES for image encryption, using statistically independent sampling and standardised datasets. The study identifies whether quantitative metrics always reflect visual encryption quality.

2. Texture Features Used

None. No texture-based or content-adaptive processing is applied. The study treats images as flat bitstreams for XOR-based encryption.

3. Encryption Algorithm

Binary key generation from chaotic trajectories followed by XOR masking of the image bitstream. Decryption repeats the same XOR with the identical key.

4. Chaotic Maps / Cryptographic Methods

Henon Map (2D), Tent Map (1D), Logistic Map (1D), AES (baseline comparison).

5. Dataset / Images Used

100 RGB images of size 512x512 from Kaggle datasets, spanning five object classes (albatross, airliner, admiral butterfly, African elephant, American lobster).

6. Security Metrics

MSE: 105-133, PSNR: 26-28 dB, NPCR: 45-54% (bit-level), UACI: 23-28% (bit-level). Note: values below standard thresholds because paper measures bit-level change rate, not pixel-level differential metric. No entropy or correlation analysis performed.

7. Experimental Results

Tent and Logistic maps achieve strong visual encryption; Henon produces partially interpretable ciphertext despite better MSE/PSNR. AES shows most uniform histograms and fastest execution. Clear disconnect between numerical metrics and visual quality.

8. Strengths

Rigorous statistical independence framework. Honest acknowledgement that standard metrics (PSNR, MSE, NPCR) can be misleading. Reproducible benchmark using standardised datasets.

9. Limitations

No entropy, correlation, or key-space analysis. Simple XOR-only encryption with no permutation/diffusion stages. No content-adaptive or region-based encryption strategy.

10. Research Gap

Proves that standard quantitative metrics do NOT guarantee encryption quality, but offers no solution. No spatial or textural adaptation - the entire image is encrypted uniformly. Motivates the need for content-aware encryption that adapts to local image complexity.

Paper 2: Multi-Image Encryption via Chaotic PRNG and DWT Compression

Authors: Y. Xu, S. Gao, Y. Cao & J. Mou (2026)

Journal: ACM Trans. Multimedia Comput. Commun. Appl., Vol. 22, No. 3

1. Research Objective

To design an efficient multi-image encryption scheme that compresses colour images via DWT before encryption, reducing resource consumption during network transmission while maintaining high security.

2. Texture Features Used

DWT (Discrete Wavelet Transform) for frequency-band decomposition and compression (LL subband at 1/4 size). No spatial texture classification or block-level adaptation.

3. Encryption Algorithm

DWT compresses each image to 1/4 size. Compressed images merged into plaintext cube. Confusion via 3D position scrambling. Diffusion via pixel-wise XOR with chaotic sequences.

4. Chaotic Maps / Cryptographic Methods

Three-dimensional discrete chaotic map used as pseudo-random number signal generator for both confusion and diffusion stages.

5. Dataset / Images Used

SIPI Image Database (USC): 256x256x3 images (4.1.01-4.1.07) and 512x512x3 images (4.2.01-4.2.07). Total 12 test images.

6. Security Metrics

NPCR: 99.6533% (mean), UACI: 33.4887% (mean), Entropy: 7.9994, Key Space: 2^{774} , Encryption Time: 3.25 s. Correlation near 0 in all directions.

7. Experimental Results

NPCR/UACI exceed standard thresholds. Entropy close to 8. Robust against noise and shearing attacks. However, PSNR of recovered images is 28-34 dB (lossy reconstruction, not lossless).

8. Strengths

Massive key space (2^{774}). Multi-image processing improves throughput. DWT compression reduces bandwidth by 75%. Excellent statistical metrics.

9. Limitations

DWT compression is lossy - recovered images NOT identical to originals (PSNR 28-34 dB). Extra space padded with zeros. No content-adaptive encryption.

10. Research Gap

Encrypts all image regions uniformly after frequency decomposition. No block-level texture analysis - smooth regions receive the same heavy encryption as complex textures, resulting in unnecessary computational overhead.

Paper 4: Texture-Adaptive Hierarchical Encryption for Remote Sensing

Authors: J. Tang, X. Jiang, C. Huang et al. (2025)

Journal: Remote Sensing, 17, 2940

1. Research Objective

To balance high security and computational efficiency for large-scale remote sensing image encryption by classifying image blocks based on texture complexity and applying hierarchical chaotic encryption accordingly.

2. Texture Features Used

Information Entropy used to classify blocks into Simple, Medium, and Complex categories. Classification based on predefined entropy thresholds with empirical prior knowledge. No ML classifier used.

3. Encryption Algorithm

Spatial partitioning into blocks. Simple blocks: 1D Logistic-Tent (lightweight XOR). Medium blocks: 2D LASM (sorting + XOR). Complex blocks: 4D Hyperchaotic (CMT scrambling + XOR). Final global XOR pass to eliminate inter-block correlation.

4. Chaotic Maps / Cryptographic Methods

1D Logistic-Tent System (LTS) for simple blocks, 2D Logistic-adjusted-Sine Map (LASM) for medium, 4D Hyperchaotic system for complex, SHA-based key derivation.

5. Dataset / Images Used

High-resolution remote sensing images: 256x256 to 4096x4096 (from figshare).

6. Security Metrics

NPCR: 99.60-99.62%, UACI: 33.45-33.54%, Entropy: 7.997-7.999, Key Space: $\geq 10^{60}$, Enc. Time (4096x4096): 9014 ms. Correlation near 0.

7. Experimental Results

All NPCR/UACI exceed thresholds. Significant efficiency gains for large images (9014 ms vs 32163 ms for AES competitor). Lossless decryption. Uniform histograms.

8. Strengths

Direct block-level texture adaptation. Excellent scalability via parallel processing. Superior efficiency for large-scale data. Global diffusion removes block artefacts.

9. Limitations

Texture classification uses predefined entropy thresholds - not universally optimal. Auxiliary metadata must be securely transmitted. Designed only for remote sensing. No ML-based classification.

10. Research Gap

Uses ONLY information entropy as the single texture metric. Entropy alone cannot capture directional textures, edge patterns, or multi-scale spatial features (GLCM). The paper itself acknowledges future work should use ML to classify blocks - exactly what our system implements.

Paper 5: OptiSecure-3D - Compressed Encryption via SAE + Optimized 3D Chaos

Authors: A. Tiwari, P. Diwan, T.D. Diwan et al. (2025)

Journal: Scientific Reports, 15:14151

1. Research Objective

To develop a compressed image encryption algorithm using a Stacked Autoencoder (SAE) for dimensionality reduction and Differential Evolution (DE) to optimise 3D Arnold Cat Map parameters for maximum security.

2. Texture Features Used

High-dimensional latent features extracted by SAE. These are compression features, not spatial texture features. No block-level texture analysis (GLCM, entropy, edge density) is performed.

3. Encryption Algorithm

SAE compresses input to latent space. 3D Arnold Cat Map generates chaotic permutation/diffusion sequences. DE optimises ACM parameters via multi-objective fitness. XOR-based diffusion. Asymmetric encryption for key exchange.

4. Chaotic Maps / Cryptographic Methods

3D Arnold Cat Map (primary engine), Differential Evolution (metaheuristic optimiser), Stacked Autoencoder (deep learning compression).

5. Dataset / Images Used

Set5, Set14, Brain MRI, Eye Fundus (Kaggle), Lena, Butterfly. Sizes: 128x128 to 1024x1024.

6. Security Metrics

NPCR: 99.43-99.89% (mean ~99.80%), UACI: ~33.46%, Entropy: 7.9984, Correlation: 0.005, Encryption Time: 0.15 s.

7. Experimental Results

NPCR 99.89% statistically significant ($z > 1.96$ at 95%). Outperforms five state-of-the-art methods. Strong noise resilience. Consistent low correlation.

8. Strengths

Evolutionary optimisation guarantees near-optimal chaotic parameters. SAE reduces data volume. Comprehensive evaluation. Fast encryption (0.15 s).

9. Limitations

SAE requires pre-training and GPU - heavy overhead. SAE compression is lossy. No block-level spatial adaptation. Challenges for dynamic multimedia.

10. Research Gap

Replaces spatial texture analysis with deep learning. Entire image treated uniformly - no region differentiation. Heavy SAE + DE pipeline requires GPU, making it impractical for resource-constrained deployments. No lightweight texture classification explored.

Paper 6: SCD-CHAOS - Dynamic S-Box Hybrid Encryption for IoT

Authors: Yogi, Majumdar, Ghosh et al. (2026)

Journal: Journal of Supercomputing (Springer)

1. Research Objective

To design a lightweight, real-time chaotic encryption scheme for IoT/edge environments using a hybrid dual-map system with dynamic S-box substitution.

2. Texture Features Used

None for block classification. Entropy used only for S-box generation, not content-adaptive encryption. All regions receive identical processing.

3. Encryption Algorithm

Three-stage pipeline: Permutation (Tent Map index reordering), Diffusion (Henon Map XOR masking), Confusion (Dynamic S-box substitution from hybrid key). Operates on 64x64 RGB images.

4. Chaotic Maps / Cryptographic Methods

Tent Map (permutation), Henon Map (diffusion), Hybrid Key (combined sequences for S-box), SHA-256 dynamic seed.

5. Dataset / Images Used

SIPI Image Database: Peppers, Tree, House, Airplane. All downsampled to 64x64. IoT: ESP32-WROOM-32, ARM Cortex-M4.

6. Security Metrics

NPCR: 99.6226% (mean), UACI: 33.4732% (mean), Entropy: 7.9991, MSE: 6904.83, SSIM (enc): 0.00195, Key Space: 2^{512} , Enc. Time (64x64): 0.006 s.

7. Experimental Results

All metrics exceed thresholds. Near-perfect entropy. SSIM near 0 for encrypted, 0.9994 for decrypted. Deployed on ESP32/STM32 with <0.5 W power. NIST tests passed.

8. Strengths

Extremely fast (0.006 s for 64x64). Low memory (480-520 KB). Massive key space (2^{512}). Dynamic S-box enhances avalanche effect. Validated on real IoT hardware.

9. Limitations

Images must be downsampled to 64x64 - severe resolution sacrifice. Not practical for medical/satellite imagery. No content-adaptive encryption.

10. Research Gap

Achieves real-time performance only by drastically sacrificing resolution (64x64). Impractical for any domain requiring high-resolution encrypted images. No block-level texture analysis - all regions receive identical encryption.

Paper 7: Multi-Party RDH in Medical Images via TGHQC

Authors: W. Wan, Y. Wang, Z. Gu et al. (2025)

Journal: J. King Saud University - Computer and Info. Sciences, 37:29

1. Research Objective

To design a multi-party reversible data hiding scheme for medical images with role-based access control (patient, hospital, doctor) while maximising embedding capacity through texture-guided encoding.

2. Texture Features Used

Edge Pixel Density via Canny Operator. Blocks classified as: PE/Smooth (density ≤ 0.1), MMD/Moderate ($0.1 < \text{density} \leq 0.2$), MND/Textured (density > 0.2). Drives differentiated encoding, not encryption strength.

3. Encryption Algorithm

TGHQC classifies blocks and applies adaptive difference generation + hierarchical encoding. Shamir's Secret Sharing for polynomial-based (t,n) threshold sharing. Dual-level embedding. Lagrange interpolation for recovery.

4. Chaotic Maps / Cryptographic Methods

Shamir's Secret Sharing (polynomial-based). No chaotic maps used for encryption. MED predictor for PE block differences.

5. Dataset / Images Used

COVID-CT dataset (4 medical images, 512x512). Classic standard images. BOSSBase and BOWS-2 for benchmarking.

6. Security Metrics

PSNR (shares): 9.56-10.12 dB, SSIM (shares): 0.008-0.014, Recovered: PSNR=infinity, SSIM=1.000, Embedding Rate: 3.65-6.37 bpp. NPCR/UACI/Entropy: Not evaluated.

7. Experimental Results

Perfect lossless reconstruction. Shares appear as noise. Higher embedding rates than 5 competing methods. Average 4.5433 bpp on medical dataset. Lower algorithmic complexity.

8. Strengths

Complete lossless recovery. Multi-party access control. Texture-guided encoding maximises capacity. No data expansion. Separable image recovery and data extraction.

9. Limitations

Focused on data hiding, NOT encryption. No NPCR/UACI/entropy metrics. No chaotic map integration. Lower payload vs dedicated encryption. Single-feature texture classification (Canny only).

10. Research Gap

Demonstrates value of texture-guided classification but applies it only to data embedding, not encryption strength. Canny edge density is a single feature - cannot capture full textural complexity of multi-feature approaches (GLCM). No chaotic encryption used.

Consolidated Research Gaps Across All Papers

Gap 1: No Spatial Texture Adaptation (Papers 1, 2, 5, 6)

Most schemes apply identical encryption globally. Smooth regions receive the same heavy computation as complex textures, wasting resources and missing efficiency-security trade-offs.

Gap 2: Single-Metric Texture Classification (Papers 4, 7)

Paper 4 uses only information entropy; Paper 7 uses only edge density. Neither captures multi-directional textural richness available through GLCM-based multi-feature analysis.

Gap 3: No ML-Based Adaptive Classification (ALL Papers)

None of the six papers use a trained ML classifier (Random Forest, SVM) to automatically route image blocks to different encryption tiers based on multi-feature texture descriptors.

Gap 4: Resolution Sacrifice for Speed (Paper 6)

Real-time IoT performance achieved only by downscaling to 64x64 - impractical for high-resolution medical, satellite, or surveillance applications.

Gap 5: Lossy Reconstruction (Papers 2, 5)

DWT and SAE compression introduce irreversible quality loss - unacceptable for medical/forensic domains requiring pixel-perfect fidelity.

Gap 6: Heavy Computational Overhead (Paper 5)

Deep learning (SAE + Differential Evolution) requires GPU hardware and pre-training, limiting real-time and embedded deployment feasibility.

How Our 3-Tier Adaptive System Addresses Every Gap

[OK] Gap 1 Addressed

GLCM-based block-level texture analysis adapts encryption strength per region: Smooth -> Logistic Map, Medium -> Lorenz System, Rough -> Hyperchaotic Chen Map.

[OK] Gap 2 Addressed

Uses 5+ texture features (GLCM contrast, dissimilarity, homogeneity, energy, correlation + local entropy) - far richer than entropy-only or edge-density-only classification.

[OK] Gap 3 Addressed

Trained Random Forest classifier automatically routes blocks to the appropriate encryption tier with no manual thresholds required.

[OK] Gap 4 Addressed

Encrypts full-resolution images (256x256, 512x512+) with no downscaling required.

[OK] Gap 5 Addressed

100% lossless decryption verified across all test images (PSNR = infinity, SSIM = 1.0).

[OK] Gap 6 Addressed

Lightweight Random Forest + NumPy vectorisation - no GPU needed, runs on standard consumer hardware.